

Brady Bangasser

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Compiler-focused engineer working close to the metal: LLVM IR optimization passes, toolchain performance, and language design, applied to real high-performance workloads.

EXPERIENCE

Iowa State University

Aug 2025 – Present

Research Assistant — Compiler Optimization

- Built OMPRTF (OpenMP RunTime Fabric), a dynamic-analysis pipeline that applies aggressive, conditional optimizations to OpenMP GPU-offload programs by lowering them to LLVM IR.
- Wrote an LLVM optimization pass that breaks opaque OpenMP runtime calls into discrete IR instructions, pruning redundant host-GPU data movement without destabilizing kernels.
- Built a profiling library that correlates OpenMP target logs with IR metadata to surface duplicate transfers and unused memory automatically.
- Co-developed an LLVM-based tool that restructures disk I/O access patterns to speed up data loading for large ML/AI workloads.

PROJECTS

Omni HTTP Router

2025–2026

- Built a Rust HTTP router that resolves middleware at compile time into fully static stacks, removing traditional routing overhead.
- Wrote a compiler that lowers a routing DSL directly to native object files, and compiles matching Terraform and container deployments alongside the app binary.

Minnesotan Programming Language

2023–2025

- Designed a Turing-complete esoteric language and built an end-to-end compiler with Flex/Bison that emits LLVM IR, compiling to native executables.
- Implemented a custom lexer and parser that transform the language's colloquial syntax into an AST before LLVM IR emission.
- Wrote the grammar and documentation so others can author and compile Minnesotan programs.

Public Data Scraping and Processing

2025

- Wrote a compiler-like tool that pairs lexical analysis with an LLM to standardize and validate data formats, and trained an ML model to flag anomalies in the collected data.

EDUCATION

Iowa State University

Ames, IA

M.S. Computer Science · GPA: 4.00

Aug 2025 – May 2027

- Concurrent B.S./M.S. program
- Coursework: Advanced Topics in High Performance Computing, Distributed Systems and Middleware, Data Security in Machine Learning, Network Design and Security

Iowa State University

Ames, IA

B.S. Computer Science · GPA: 3.79 · Dean's List

Aug 2024 – Dec 2026

- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Rust, Bash, CUDA, Lua

Compilers & HPC: LLVM, Flex/Bison, OpenMP